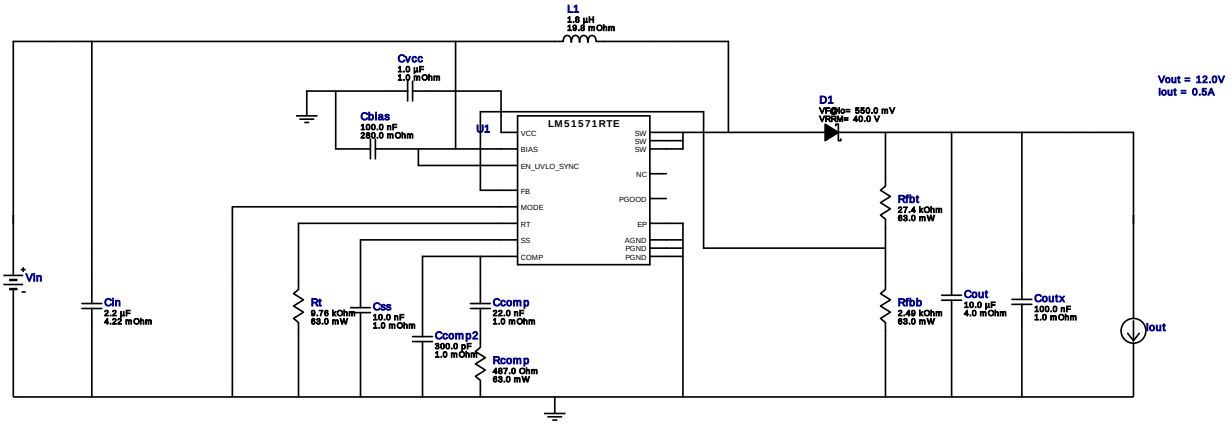
VinMin = 3.0V
VinMax = 5.0V
Vout = 12.0V
Iout = 0.5A

Device = LM51571RTE
Topology = Boost
Created = 2026-10-13 07:01:11.201
BOM Cost = $1.38
BOM Count = 15
Total Pd = 0.88W

WEBENCH® Design Report

Design : 4 LM51571RTE
LM51571RTE 3V-5V to 12.00V @ 0.5A

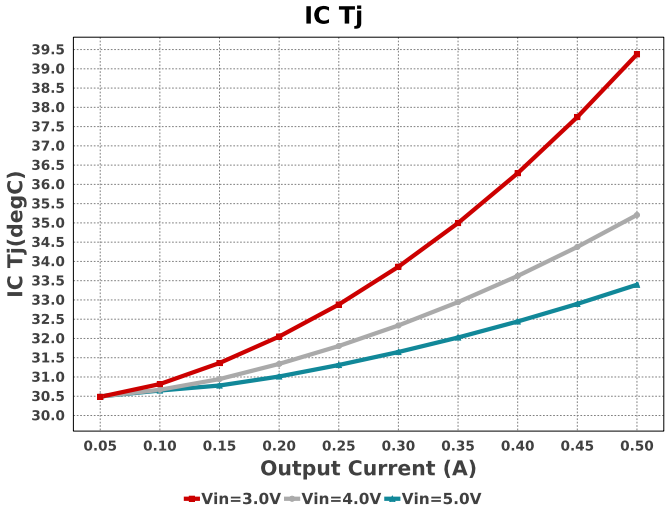


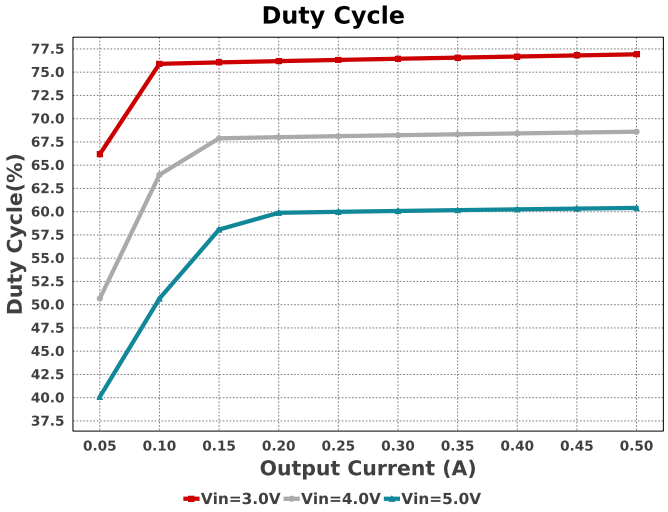
Electrical BOM

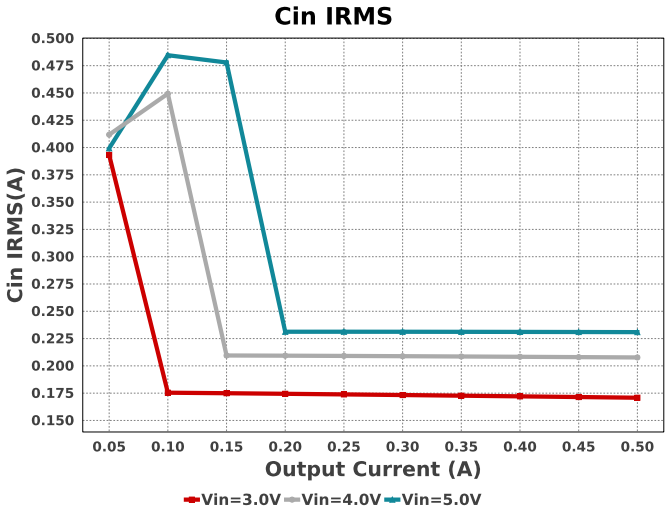
Name	Manufacturer	Part Number	Properties	Qty	Price	Footprint
Cbias	AVX	08053C104KAT2A Series= X7R	Cap= 100.0 nF ESR= 280.0 mOhm VDC= 25.0 V IRMS= 0.0 A	1	$0.01	0805 7 mm²
Ccomp	MuRata	GRM155R71C223KA01D Series= X7R	Cap= 22.0 nF ESR= 1.0 mOhm VDC= 16.0 V IRMS= 0.0 A	1	$0.01	0402 3 mm²
Ccomp2	MuRata	GRM1555C1E301JA01D Series= C0G/NP0	Cap= 300.0 pF ESR= 1.0 mOhm VDC= 25.0 V IRMS= 0.0 A	1	$0.02	0402 3 mm²
Cin	MuRata	GRM21BR71A225KA01L Series= X7R	Cap= 2.2 uF ESR= 4.22 mOhm VDC= 10.0 V IRMS= 2.08454 A	1	$0.03	0805 7 mm²
Cout	MuRata	GRM31CR71E106KA12L Series= X7R	Cap= 10.0 uF ESR= 4.0 mOhm VDC= 25.0 V IRMS= 6.0 A	1	$0.06	1206_180 11 mm²
Coutx	MuRata	GRM155R71C104KA88D Series= X7R	Cap= 100.0 nF ESR= 1.0 mOhm VDC= 16.0 V IRMS= 0.0 A	1	$0.01	0402 3 mm²
Css	MuRata	GRM033R70J103KA01D Series= X7R	Cap= 10.0 nF ESR= 1.0 mOhm VDC= 6.3 V IRMS= 0.0 A	1	$0.01	0201 2 mm²
Cvcc	Taiyo Yuden	EMK107B7105KA-T Series= X7R	Cap= 1.0 uF ESR= 1.0 mOhm VDC= 16.0 V IRMS= 0.0 A	1	$0.01	0603 5 mm²
D1	Fairchild Semiconductor	SS14FL	VF@Io= 550.0 mV VRRM= 40.0 V	1	$0.03	SOD-123F 12 mm²

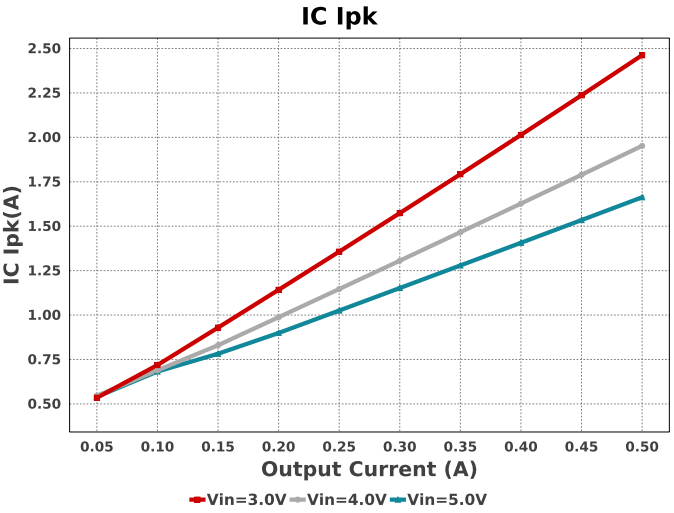
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Name	Manufacturer	Part Number	Properties	Qty	Price	Footprint
L1	Bourns	SRN6045-1R8Y	L= 1.8 µH 19.8 mOhm	1	$0.25	 SRN6045 64 mm²
Rcomp	Vishay-Dale	CRCW0402487RFKED Series= CRCW..e3	Res= 487.0 Ohm Power= 63.0 mW Tolerance= 1.0%	1	$0.01	 0402 3 mm²
Rfbb	Vishay-Dale	CRCW04022K49FKED Series= CRCW..e3	Res= 2.49 kOhm Power= 63.0 mW Tolerance= 1.0%	1	$0.01	 0402 3 mm²
Rfbt	Vishay-Dale	CRCW040227K4FKED Series= CRCW..e3	Res= 27.4 kOhm Power= 63.0 mW Tolerance= 1.0%	1	$0.01	 0402 3 mm²
Rt	Vishay-Dale	CRCW04029K76FKED Series= CRCW..e3	Res= 9.76 kOhm Power= 63.0 mW Tolerance= 1.0%	1	$0.01	 0402 3 mm²
U1	Texas Instruments	LM51571RTE	Switcher	1	$0.90	RTE0016K 16 mm²



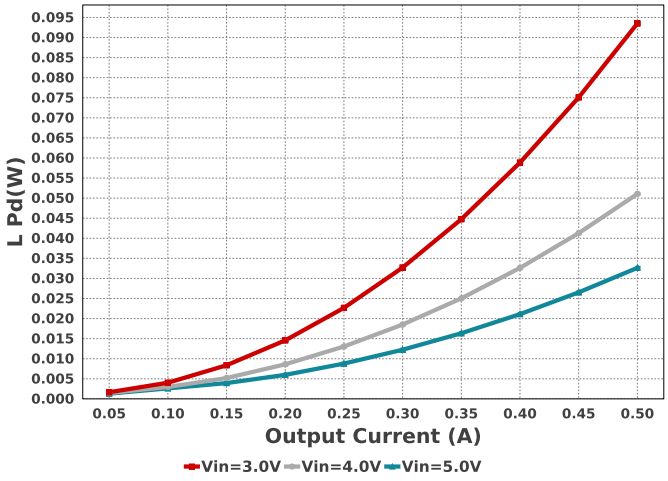




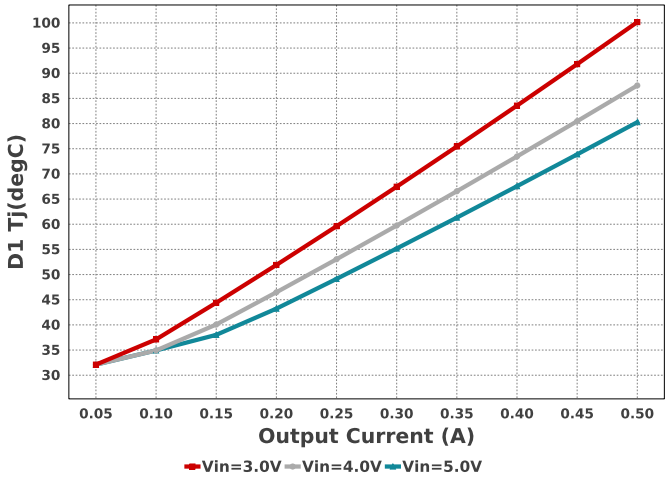


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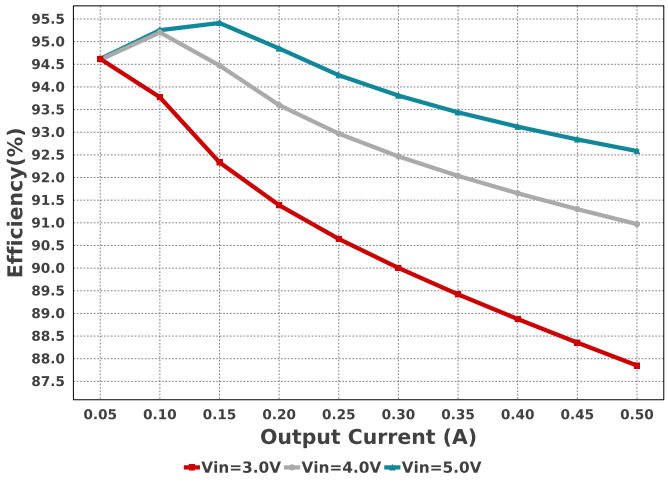
L Pd



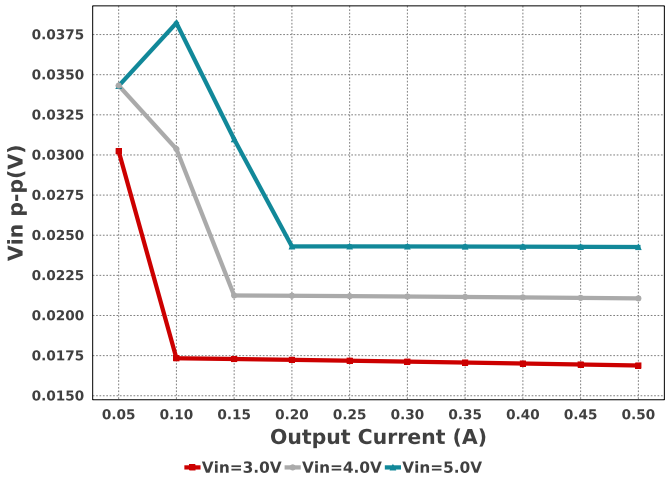
D1 Tj



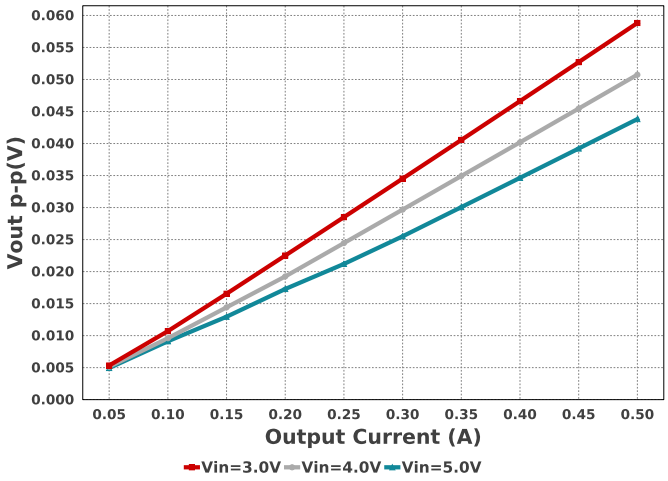
Efficiency



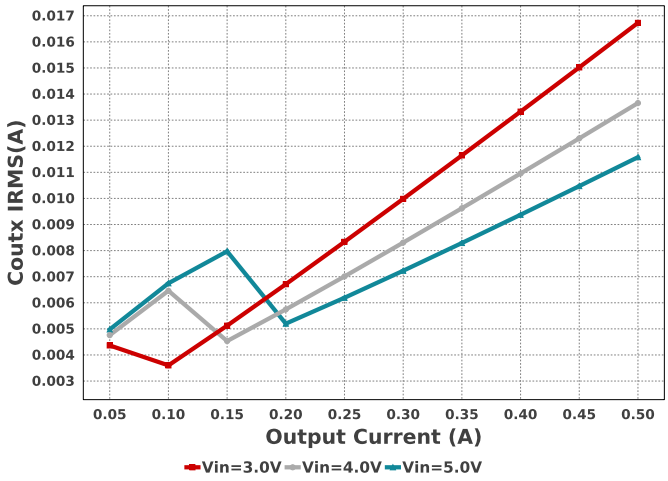
Vin p-p



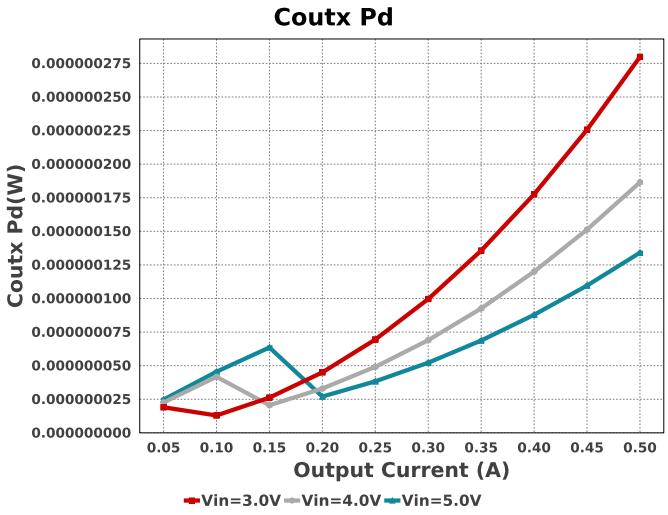
Vout p-p

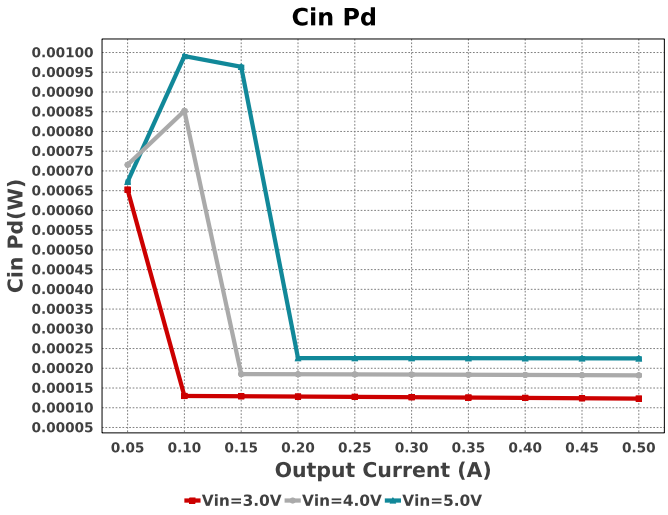


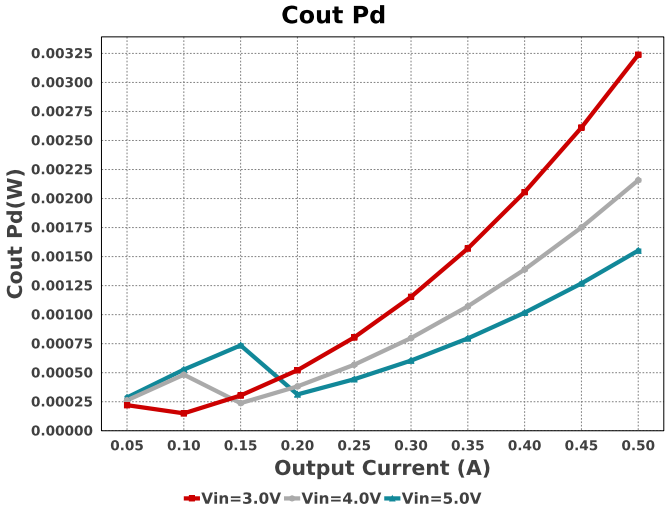
Coutx IRMS

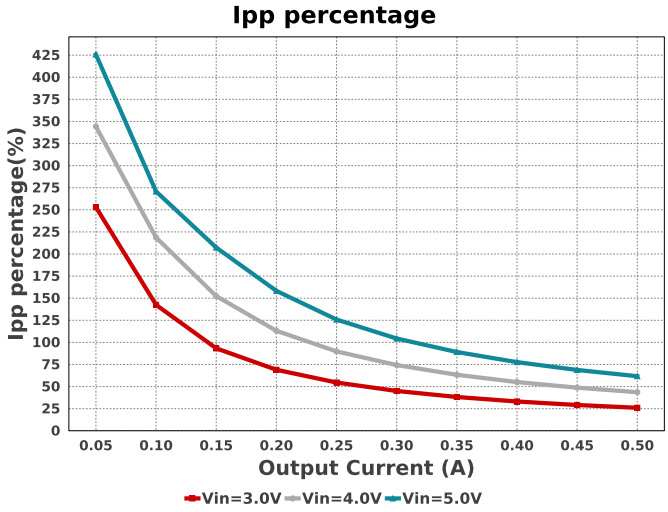


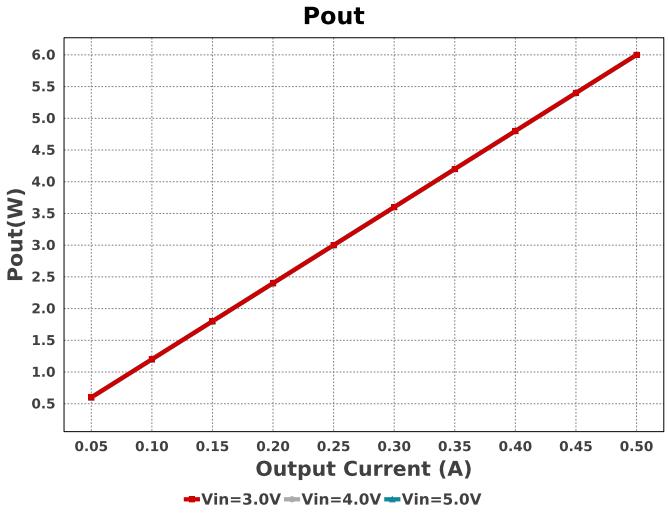
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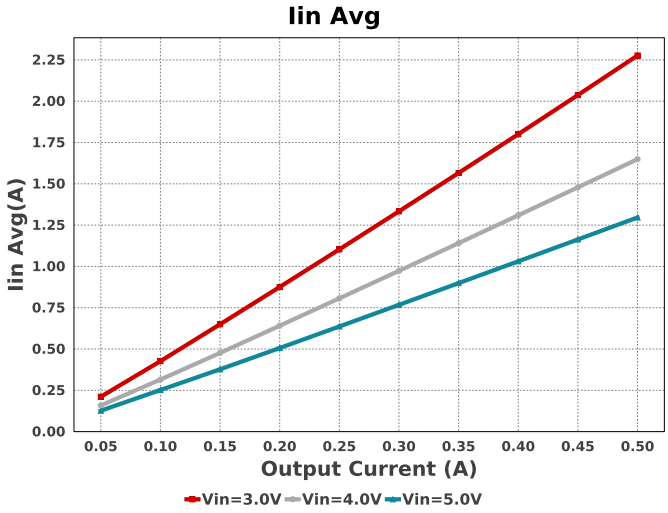




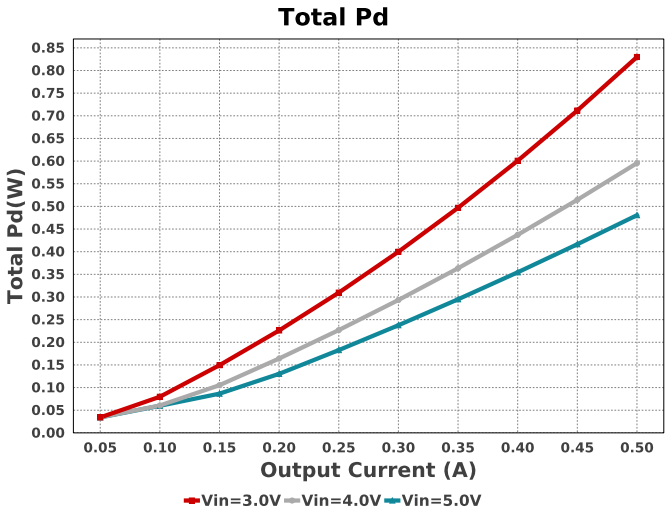


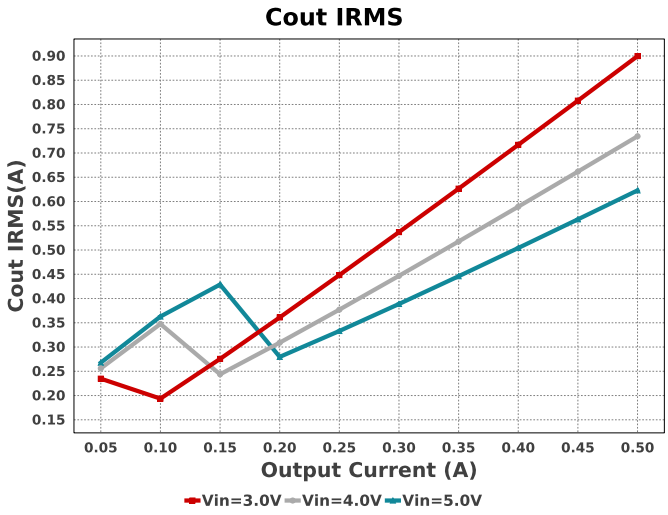


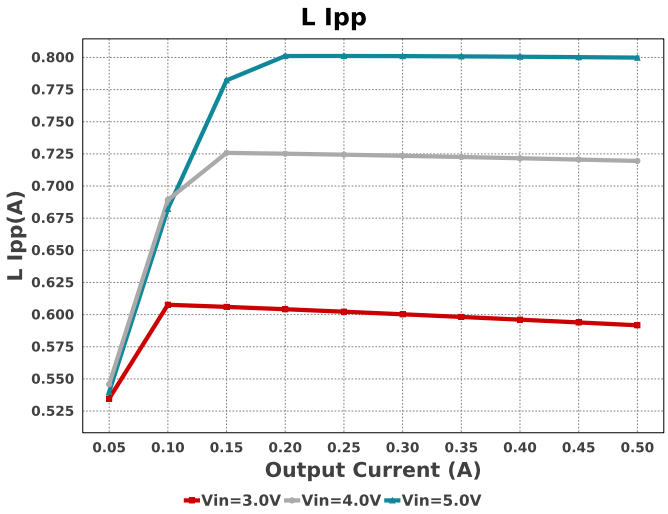


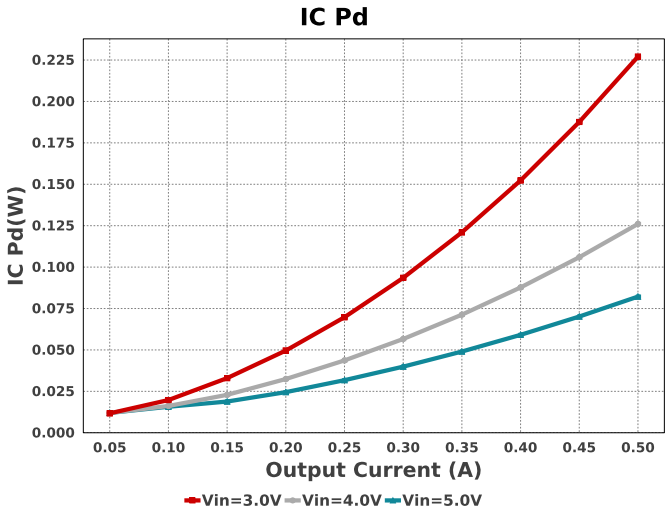


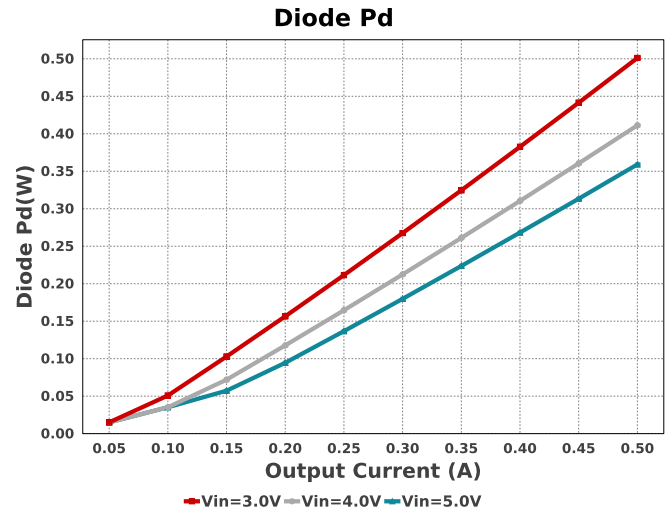
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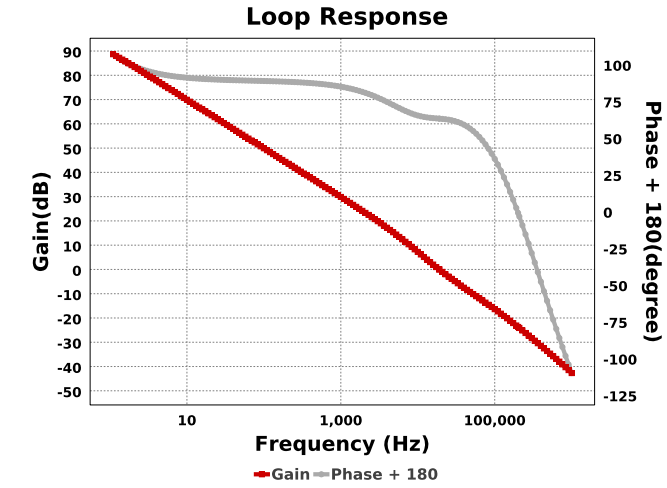












Operating Values

#	Name	Value	Category	Description
1.	Cin IRMS	171.101 mA	Capacitor	Input capacitor RMS ripple current
2.	Cin Pd	123.54 µW	Capacitor	Input capacitor power dissipation
3.	Cout IRMS	920.514 mA	Capacitor	Output capacitor RMS ripple current
4.	Cout Pd	3.389 mW	Capacitor	Output capacitor power dissipation
5.	D1 Tj	106.15 degC	Diode	D1 junction temperature
6.	Diode Pd	543.92 mW	Diode	Diode power dissipation
7.	IC Ipk	2.478 A	IC	Peak switch current in IC
8.	IC Pd	230.45 mW	IC	IC power dissipation
9.	IC Tj	39.518 degC	IC	IC junction temperature
10.	IC Tolerance	10.0 mV	IC	IC Feedback Tolerance
11.	ICThetaJA	41.3 degC/W	IC	IC junction-to-ambient thermal resistance

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#	Name	Value	Category	Description
12.	Iin Avg	2.292 A	IC	Average input current
13.	Ipp percentage	25.854 %	Inductor	Inductor ripple current percentage (with respect to average inductor current)
14.	L Ipp	592.71 mA	Inductor	Peak-to-peak inductor ripple current
15.	L Pd	94.787 mW	Inductor	Inductor power dissipation
16.	Cin Pd	123.54 µW	Power	Input capacitor power dissipation
17.	Cout Pd	3.389 mW	Power	Output capacitor power dissipation
18.	Diode Pd	543.92 mW	Power	Diode power dissipation
19.	IC Pd	230.45 mW	Power	IC power dissipation
20.	L Pd	94.787 mW	Power	Inductor power dissipation
21.	Total Pd	877.5 mW	Power	Total Power Dissipation
22.	BOM Count	15	System Information	Total Design BOM count
23.	Cross Freq	13.381 kHz	System Information	Bode plot crossover frequency
24.	Duty Cycle	77.078 %	System Information	Duty cycle
25.	Efficiency	87.241 %	System Information	Steady state efficiency
26.	FootPrint	144.0 mm^2	System Information	Total Foot Print Area of BOM components
27.	Frequency	2.063 MHz	System Information	Switching frequency
28.	Gain Marg	-20.103 dB	System Information	Bode Plot Gain Margin
29.	Iout	500.0 mA	System Information	Iout operating point
30.	Low Freq Gain	87.695 dB	System Information	Gain at 1Hz
31.	Mode	CCM	System Information	Conduction Mode
32.	Phase Marg	51.665 deg	System Information	Bode Plot Phase Margin
33.	Pout	6.0 W	System Information	Total output power
34.	Total BOM	$1.38	System Information	Total BOM Cost
35.	Vin	3.0 V	System Information	Vin operating point
36.	Vin p-p	16.914 mV	System Information	Input Source ripple voltage
37.	Vout	12.0 V	System Information	Operational Output Voltage
38.	Vout Actual	12.004 V	System Information	Vout Actual calculated based on selected voltage divider resistors
39.	Vout Tolerance	2.87 %	System Information	Vout Tolerance based on IC Tolerance (no load) and voltage divider resistors if applicable
40.	Vout p-p	60.083 mV	System Information	Peak-to-peak output ripple voltage

Design Inputs

Name	Value	Description
Iout	500.0 m	Maximum Output Current
VinMax	5.0	Maximum input voltage
VinMin	3.0	Minimum input voltage
Vout	12.0	Output Voltage
base_pn	LM51571	Base Product Number
source	DC	Input Source Type
Ta	30.0	Ambient temperature

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WEBENCH® Assembly

Component Testing

Some published data on components in datasheets such as Capacitor ESR and Inductor DC resistance is based on conservative values that will guarantee that the components always exceed the specification. For design purposes it is usually better to work with typical values. Since this data is not always available it is a good practice to measure the Capacitance and ESR values of Cin and Cout, and the inductance and DC resistance of L1 before assembly of the board. Any large discrepancies in values should be electrically simulated in WEBENCH to check for instabilities and thermally simulated in WebTHERM to make sure critical temperatures are not exceeded.

Soldering Component to Board

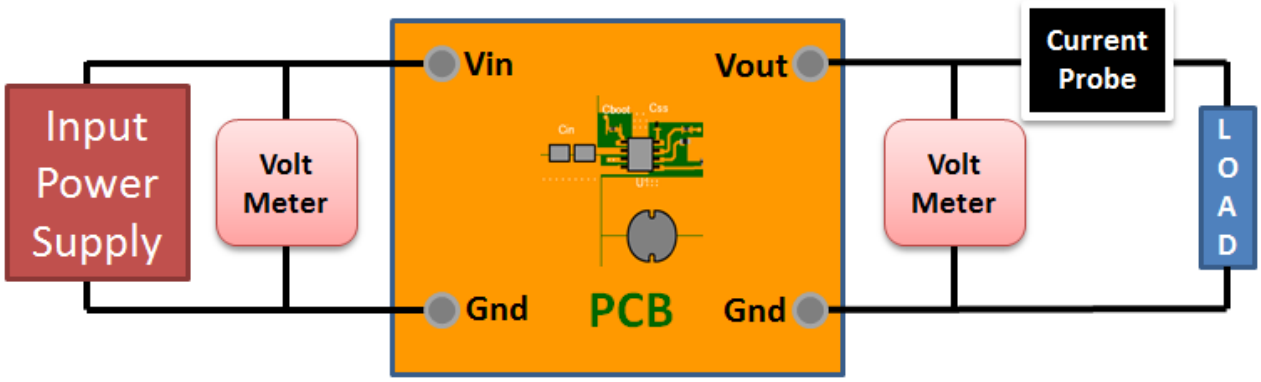
If board assembly is done in house it is best to tack down one terminal of a component on the board then solder the other terminal. For surface mount parts with large tabs, such as the DPAK, the tab on the back of the package should be pre-tinned with solder, then tacked into place by one of the pins. To solder the tab town to the board place the iron down on the board while resting against the tab, heating both surfaces simultaneously. Apply light pressure to the top of the plastic case until the solder flows around the part and the part is flush with the PCB. If the solder is not flowing around the board you may need a higher wattage iron (generally 25W to 30W is enough).

Initial Startup of Circuit

It is best to initially power up the board by setting the input supply voltage to the lowest operating input voltage 3.0V and set the input supply's current limit to zero. With the input supply off connect up the input supply to Vin and GND. Connect a digital volt meter and a load if needed to set the minimum Iout of the design from Vout and GND. Turn on the input supply and slowly turn up the current limit on the input supply. If the voltage starts to rise on the input supply continue increasing the input supply current limit while watching the output voltage. If the current increases on the input supply, but the voltage remains near zero, then there may be a short or a component misplaced on the board. Power down the board and visually inspect for solder bridges and recheck the diode and capacitor polarities. Once the power supply circuit is operational then more extensive testing may include full load testing, transient load and line tests to compare with simulation results.

Load Testing

The setup is the same as the initial startup, except that an additional digital voltmeter is connected between Vin and GND, a load is connected between Vout and GND and a current meter is connected in series between Vout and the load. The load must be able to handle at least rated output power + 50% (7.5 watts for this design). Ideally the load is supplied in the form of a variable load test unit. It can also be done in the form of suitably large power resistors. When using an oscilloscope to measure waveforms on the prototype board, the ground leads of the oscilloscope probes should be as short as possible and the area of the loop formed by the ground lead should be kept to a minimum. This will help reduce ground lead inductance and eliminate EMI noise that is not actually present in the circuit.



Design Assistance

1. Master key : C0723DD09D79F8AFBAB58B2D8AAED5F5[v1]
2. **LM51571** Product Folder : http://www.ti.com/product/LM51571 : contains the data sheet and other resources.

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